

# Notes: Holmström and Milgrom (AER, 1994)

## The Firm as an Incentive System

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# 1 Big picture and incremental contribution

- **Question.** Why do employment relationships and independent-contractor relationships differ along many dimensions at once: compensation, asset ownership, monitoring, job restrictions, freedom to serve other principals, and control rights?
- **Core idea.** Employment and contracting are not isolated contractual clauses. They are *incentive systems*. Performance pay, asset ownership, and worker freedom are jointly chosen instruments, and their optimal use depends on complementarities across a worker’s multiple tasks.
- **Main theoretical claim.** High-powered incentives, asset ownership by the worker, and freedom from direct controls are complementary when the worker’s activities compete for attention and when incentives for one task can distort effort away from other tasks.
- **Main empirical claim.** In settings such as industrial selling, the instruments should cluster: independent representatives should have high commissions, ownership or effective control of customer relationships, and freedom to sell other firms’ products; employee sales agents should have lower commissions, firm ownership of customer relationships, and tighter restrictions.
- **Key methodological contribution.** The paper shows that complementarity of instruments is not enough to imply positive correlation in cross-sectional data. One must also know which *exogenous parameters* move all instruments in the same direction and how the parameters are distributed.
- **Statistical contribution.** The paper uses supermodularity, associated random variables, and affiliated random variables to translate comparative statics into testable positive covariation predictions.
- **Application.** The paper applies the framework to sales-force organization, interpreting evidence from Anderson and Schmittlein and Anderson on why firms use in-house employees in some regions and independent representatives in others.

## 1.1 Incremental contribution of the paper

- **Beyond one-instrument theories of the firm.** Earlier theories emphasized one margin at a time: Coase/Simon emphasized authority, Grossman–Hart emphasized asset ownership, and principal-agent models emphasized pay-for-performance. This paper’s incremental contribution is to make these margins jointly endogenous.
- **From “agency costs” to testable systems.** The paper argues that a generic appeal to agency costs is too vague. Different agency environments imply different signs for the same empirical relationship, so researchers must tailor the incentive model to the institutional context.
- **A theory of covariation, not just comparative statics.** Standard theory often says how an endogenous variable changes when one parameter changes. This paper asks when multiple endogenous instruments should be positively correlated in observed cross sections.
- **Bridge between theory and empirics.** The paper gives conditions under which comparative statics survive unobserved heterogeneity, censored parameters, and measurement error using association and affiliation.
- **Organizational implication.** Low-powered incentives and bureaucratic restrictions are not merely defects of firms. They can be optimal tools for balancing distorted multitask incentives.
- **Management implication.** The current trend toward worker empowerment can be understood as an equilibrium response to changes in monitoring technology, task importance, and asset-return patterns, not simply as a cultural fashion.

| Incremental contribution |                                                                                               | Core tradeoff        |                                                                             |
|--------------------------|-----------------------------------------------------------------------------------------------|----------------------|-----------------------------------------------------------------------------|
| Dimension                | Contribution                                                                                  | Instrument           | Main function                                                               |
| Theory of firm           | Treats employment and contracting as bundles of complementary incentive instruments.          | Commissions          | Reward measurable output but can distort effort away from unmeasured tasks. |
| Agency theory            | Embeds asset ownership, pay, and job restrictions in one multitask model.                     | Asset ownership      | Rewards maintenance, customer nurturing, and implicit returns.              |
| Empirics                 | Explains why observed organizational attributes should covary and when they should not.       | Restrictions         | Prevent outside activities and reduce excessive market incentives.          |
| Statistics               | Uses association and affiliation to map comparative statics into cross-sectional predictions. | Monitoring precision | Determines whether narrow performance pay is useful.                        |

## 2 Model primitives and measurement

### 2.1 No empirical data set: theoretical model plus evidence-mapping

This is a theory paper. It does not construct a new data set or estimate new regressions. Its “data” contribution is instead to translate a multitask principal-agent model into empirical predictions that can be compared with existing evidence, especially on industrial sales-force organization.

**Interpretation:** The paper should be read as a theory-to-empirics bridge. Its empirical value comes from telling researchers what covariations to expect and what exogenous sources of variation must be controlled.

### 2.2 Agent tasks and performance measures

The agent chooses effort across  $N$  activities:

$$t = (t_1, \dots, t_N).$$

Effort is not directly observed. Instead, the principal observes a vector of noisy performance measures:

$$X_i = F_i(t) + \varepsilon_i, \quad i = 1, \dots, I, \quad (1)$$

where the measurement errors have mean zero.

The principal can write a linear compensation contract:

$$s(X) = \sum_i \alpha_i X_i + \beta. \quad (2)$$

The coefficients  $\alpha_i$  are commission rates or performance-pay intensities, and  $\beta$  is fixed salary.

**Interpretation:** The linear contract is inherited from the Holmström–Milgrom dynamic moral-hazard framework. In stationary environments with continuous performance feedback, the optimal dynamic contract has a tractable linear reduced form.

### 2.3 Private costs and principal benefits

The agent incurs private cost or receives private return  $C(t)$ . This term includes effort costs, opportunity costs, human-capital gains, and any private value from activities. The principal receives benefit  $B(t)$ , which can include profit, revenue, or returns to assets controlled by the principal.

**Interpretation:** The model separates private and principal payoffs. This matters because asset ownership and job restrictions can change who captures returns from an activity even if effort itself is not directly contractible.

## 2.4 Asset ownership and job restrictions

The model adds instruments beyond commissions:

- $A_j$  indicates whether the agent owns or controls asset  $j$  and therefore receives return  $G_j(t)$ .
- $\delta_k$  indicates whether the agent is allowed to undertake outside/private activity  $k$ , yielding return  $H_k(t)$ .
- $\eta$  represents the precision of performance measurement or monitoring.

**Interpretation:** These instruments correspond to common real-world distinctions between employees and contractors. Employees often receive low commissions, do not own customer relationships, and face restrictions on outside work. Contractors often receive high-powered pay, own or control customer relationships, and have more freedom.

## 2.5 Agent utility and certainty equivalent

The agent has constant absolute risk aversion:

$$U(A) = -\exp(-rA). \quad (8)$$

Because income is normally distributed, the agent's certainty equivalent is:

$$\begin{aligned} ACE(t, \alpha, A, \delta, \eta) = & \sum_i \alpha_i F_i(t) + \sum_j A_j G_j(t) + \sum_k \delta_k H_k(t) - C(t) \\ & - \frac{1}{2} r V(\alpha, A, \delta, \eta). \end{aligned} \quad (9)$$

**Interpretation:** The variance term captures incentive risk. Better measurement precision lowers the risk cost of high-powered performance pay.

## 2.6 Total certainty equivalent and organizational choice

Because fixed payments can transfer surplus between principal and agent, efficient organizational design maximizes total certainty equivalent:

$$\max TCE(t, \alpha, A, \delta, \eta), \quad (10)$$

subject to the agent choosing effort optimally. Substituting the agent's effort response  $t(\alpha, A, \delta)$  gives the organizational design problem:

$$\max T(\alpha, A, \delta, \eta). \quad (11)$$

**Interpretation:** The choice variables are not only wages. The firm chooses a system: performance pay, ownership/control rights, restrictions, and monitoring precision.

| Model objects |                                              | Employment versus contracting |                                |                          |
|---------------|----------------------------------------------|-------------------------------|--------------------------------|--------------------------|
| Object        | Meaning                                      | Instrument                    | Employment                     | Contracting              |
| $t$           | Vector of agent efforts across tasks.        | Pay                           | salary/weak commission         | high commis-sion         |
| $X_i$         | Noisy measurable performance signal.         | Asset control                 | firm owns cus-tomer/asset      | agent owns/control asset |
| $\alpha_i$    | Commission/performance-pay intensity.        |                               |                                |                          |
| $A_j$         | Agent ownership/control of asset $j$ .       | Outside activity Monitoring   | restricted direct supervi-sion | permitted outcome-based  |
| $\delta_k$    | Permission for outside/private activity.     |                               |                                |                          |
| $\eta$        | Monitoring precision or measurement quality. |                               |                                |                          |

### 3 Analytical framework

#### 3.1 Supermodularity and monotone comparative statics

The paper uses supermodularity to study how optimal incentive instruments move together. A function  $f$  is supermodular if:

$$f(z \vee z') + f(z \wedge z') \geq f(z) + f(z'). \quad (1)$$

For smooth functions, this means:

$$\frac{\partial^2 f}{\partial z_i \partial z_j} \geq 0 \quad \text{for } i \neq j. \quad (2)$$

**Interpretation:** Supermodularity formalizes complementarity. If one incentive instrument becomes more valuable, the marginal benefit of using other instruments can rise too.

#### 3.2 Theorem 1: monotone choice

If  $f(x, \pi)$  is continuous and supermodular in the choice vector  $x$  and parameter vector  $\pi$ , and the feasible set is a compact sublattice, then the largest and smallest maximizers exist and are nondecreasing functions of  $\pi$ .

**Interpretation:** This is the comparative-statics engine. If total surplus is supermodular, increasing an exogenous parameter raises all the components of the optimal incentive system.

#### 3.3 Association and affiliation

A vector of random variables  $\pi$  is associated if, for all nondecreasing functions  $f$  and  $g$ ,

$$E[f(\pi)g(\pi)] \geq E[f(\pi)]E[g(\pi)]. \quad (12)$$

Affiliation is stronger: variables remain associated conditional on sublattices.

**Interpretation:** Association lets the paper move from comparative statics to cross-sectional predictions. If parameters are associated and optimal instruments are increasing in those parameters, then the instruments are also positively associated.

#### 3.4 Theorems 2–4: from theory to empirical predictions

- **Theorem 2.** If parameters are associated, then nondecreasing transformations of them are associated. Subvectors also remain associated.

- **Theorem 3.** If a variable is monotone in associated parameters and nonmonotone only through independent nuisance parameters, association can still be preserved.
- **Theorem 4.** Association implies monotone conditional expectations over increasing sets.

**Interpretation:** These results provide the statistical logic behind testing organizational complementarities. They explain when clustered incentive attributes should appear in data even if some heterogeneity is unobserved.

## 4 Sales-force application

### 4.1 Industrial selling environment

The application is industrial selling, where firms choose between in-house employees and independent sales representatives. A salesperson has multiple competing activities:

1. direct selling effort, which can be measured through sales;
2. maintaining or nurturing customer relationships, which creates future value;
3. nonselling activities, such as servicing customers or providing information;
4. selling products for other manufacturers, which may benefit the agent but not the principal.

**Interpretation:** This is a natural multitask environment. Strong sales commissions may increase direct selling while reducing effort devoted to customer service, relationship maintenance, or firm-specific tasks.

### 4.2 The key incentive instruments

The sales-force version of the model focuses on four instruments:

- $\alpha$ : commission rate on measured sales;
- $A$ : ownership/control of customer relationships;
- $\delta$ : freedom to sell other manufacturers' products;
- $\eta$ : precision of sales-performance measurement.

**Interpretation:** Independent representation corresponds to high  $\alpha$ , high  $A$ , high  $\delta$ , and high reliance on measured sales. Employment corresponds to low  $\alpha$ , firm ownership of customer relationships, restrictions on outside activity, and more direct control.

### 4.3 Effort supply assumptions

The paper imposes two assumptions on effort supply:

1. **Task substitutability:**

$$\frac{\partial t_i}{\partial a_j} < 0 \quad \text{for } i \neq j.$$

Raising incentives for one activity draws effort away from others.

2. **Supermodular effort supply:** responsiveness to one incentive is stronger when other incentives are higher.

**Interpretation:** The first assumption creates the central problem: excessive incentives for selling can crowd out nonselling work. The second makes incentive instruments reinforce each other.

#### 4.4 Proposition 1: complementarity of instruments

Proposition 1 shows that under the effort-supply assumptions, the instruments  $\alpha$ ,  $A$ ,  $\delta$ , and  $\eta$  are complementary in the total surplus function after the sign of the nonselling-activity parameter is chosen correctly.

**Interpretation:** The proposition formalizes why a coherent incentive system should move multiple margins together. High commissions, agent ownership, and freedom from restrictions are mutually reinforcing in some environments, while low commissions, firm ownership, and restrictions are mutually reinforcing in others.

#### 4.5 Proposition 2: comparative statics

Proposition 2 states that, all else equal, increases in the following forces raise the optimal use of high-powered incentive instruments:

- lower importance of nonselling activities;
- lower variance of returns on future sales/customer relationships;
- lower variance of outside selling options;
- lower marginal cost of measuring sales performance.

Equivalently, when nonselling activities are important or sales measurement is hard, the optimal system shifts toward employment: lower commissions, firm ownership of customer relationships, and tighter restrictions on outside selling.

**Interpretation:** The prediction is not simply “incentives are complements.” It identifies the exogenous parameters that cause the whole incentive system to move toward employment or contracting.

#### 4.6 Propositions 3–5: cross-sectional predictions

- **Proposition 3.** Under affiliation of parameters and appropriate controls, commissions, customer ownership, freedom from restrictions, and measurement precision should be associated in cross-sectional data.
- **Proposition 4.** Independent agents should have higher commissions, lower job restrictions, and greater ownership/control of customer relationships than employees.
- **Proposition 5.** If the marginal cost of monitoring is independent of other parameters, then the probability of employment is nonincreasing in monitoring ease; easier measurement pushes toward contracting.

**Interpretation:** These propositions convert the theory into empirical statements. They also warn that some parameters, such as marginal benefits and costs of activities, may generate contrary movements and must be controlled.

| Predicted comovements                |                           |                                     |         | Empirical interpretation                    |                                           |
|--------------------------------------|---------------------------|-------------------------------------|---------|---------------------------------------------|-------------------------------------------|
| Exogenous change                     | Predicted system response |                                     |         | Observed pattern                            | Model interpretation                      |
| Sales measurement easier             | higher                    | commissions,                        | more    | High commissions plus agent-owned customers | independent representation                |
| Nonselling activities more important | lower                     | commissions,                        | more    | Salary plus firm-owned customers            | employment relationship                   |
| Customer-return risk lower           | more                      | customer ownership/control by agent |         | Outside-sales permission                    | high-powered external incentives          |
| Outside selling less risky           | more                      | freedom to sell products            | outside | Job restrictions                            | deliberate muting of competing incentives |

## 5 Theory-to-evidence strategy

### 5.1 Industrial sales evidence

The paper compares its predictions with the empirical studies of Anderson and Schmittlein and Anderson. Those studies find that two variables are especially important in explaining the choice between in-house employees and independent representatives:

- difficulty of evaluating performance;
- importance of nonselling activities.

The paper argues that these findings match its predictions. In-house employees have lower-powered pay and tighter restrictions, while independent representatives have stronger commissions and greater freedom.

**Interpretation:** This evidence is not estimated by the paper itself; it is used as a theory check. The important contribution is that the model predicts the signs of the cross-sectional covariations and identifies which controls matter.

### 5.2 Gasoline retailing and franchising evidence

The paper also discusses evidence from gasoline stations and franchising. Shepard and Slade find patterns that fit the theory: independent stations are more common where extra services are hard to monitor, and independent stations retain a larger fraction of gasoline revenue.

Franchising evidence is more mixed. Some franchise patterns match the model, but the relation between monitoring costs and franchising can reverse. The authors argue that this does not refute the model; rather, franchising differs because brand-name externalities make restrictions and monitoring valuable even for independent owners.

**Interpretation:** The paper's treatment of contradictory evidence is important. It rejects the idea that generic agency theory makes one universal prediction. Instead, agency models must be tailored to the activities and externalities in the specific setting.

## 6 Identification and theoretical strategy

### 6.1 What is identified?

The paper does not identify a causal effect from new data. It identifies theoretical conditions under which incentive instruments should be complements and conditions under which those complementarities should appear as positive covariation in cross-sectional evidence.

**Interpretation:** The right way to read the paper is as a modeling and empirical-design guide. It tells empiricists how to map a theory of organizational systems into testable restrictions.

## 6.2 Why supermodularity matters

Supermodularity gives monotone comparative statics without requiring linearity or concavity. It lets the model handle multiple endogenous instruments at once.

**Interpretation:** Without supermodularity, changing one instrument can have ambiguous effects on the others. With supermodularity, system-level comparative statics become tractable.

## 6.3 Why association and affiliation matter

Even if each incentive instrument is increasing in a parameter, cross-sectional covariation can fail if parameters are negatively correlated. Association and affiliation are distributional assumptions that preserve positive covariation after monotone transformations and conditioning.

**Interpretation:** This is the statistical innovation. The paper explains why comparative-statics theory is not automatically a cross-sectional prediction.

## 6.4 What must be controlled?

Some parameters move instruments in contrary directions. In the sales-force model, changes in marginal benefits of direct selling and customer nurturing can lead to negative comovements. Empirical work must control such parameters when testing the system prediction.

**Interpretation:** This is one of the paper’s most useful messages for empirical researchers: “control variables” should be chosen from the incentive-system model, not added ad hoc.

# 7 Structured summary tables

The original paper contains no numbered tables or figures. Therefore, there are no original visual assets to crop. The following compact tables are generated in these notes to summarize the paper’s theoretical structure. Closely related tables are placed on the same row; any standalone table is placed at width 0.6, following the requested format rule.

| Model modules   |                                                                | Main predictions           |                                                        |
|-----------------|----------------------------------------------------------------|----------------------------|--------------------------------------------------------|
| Module          | Purpose                                                        | Condition                  | Prediction                                             |
| General model   | Embeds commissions, asset ownership, and restrictions.         | Sales easy to measure      | high commissions, customer ownership, outside freedom. |
| Supermodularity | Shows when instruments are complements.                        | Nonselling tasks important | low-powered pay, restrictions, firm ownership.         |
| Association     | Translates monotone choices into statistical covariation.      | Instrument complementarity | employment and contracting attributes cluster.         |
| Sales model     | Applies the logic to employee versus independent sales agents. | Troublesome parameters     | need empirical controls.                               |

Why there are no original figures/tables

| Item              | Explanation                      |
|-------------------|----------------------------------|
| Original figures  | None in the article.             |
| Original tables   | None in the article.             |
| Visual extraction | No paper figures/tables to crop. |
| Notes visuals     | Generated summary tables only.   |

## 8 Interpretive synthesis

### 8.1 Employment as balanced low-powered incentives

Employment is not simply weak incentives. It is a coherent system in which weak commissions, firm asset ownership, and restrictions on outside activity jointly prevent the agent from overinvesting in narrowly rewarded tasks and neglecting hard-to-measure activities.

### 8.2 Contracting as balanced high-powered incentives

Independent contracting is also a coherent system. High commissions work better when the worker owns or controls relevant assets and has freedom to exploit opportunities outside the principal's direct control.

### 8.3 Complementarities do not automatically create correlations

A central lesson is methodological: the fact that instruments are complements in the objective does not imply that they will be positively correlated in data. Researchers must know which exogenous parameters move the system and whether those parameters are associated.

### 8.4 Why low-powered incentives can be good

Low-powered incentives are often criticized as bureaucratic inefficiencies. The paper shows that they can be optimal when high-powered incentives would distort effort across multiple tasks. Muting one incentive can improve total performance by balancing the worker's task portfolio.

### 8.5 Why agency models must be context-specific

The paper's discussion of franchising shows that the same agency paradigm can imply different predictions in different contexts. Industrial sales and franchising have different externalities, so one should not expect a single monitoring-cost prediction to apply everywhere.

## 9 Short summary

- The paper explains employment and independent contracting as alternative incentive systems.
- The key instruments are performance pay, asset ownership, and job restrictions.
- In multitask environments, high-powered incentives for one activity can distort effort away from other activities.
- The optimal system balances incentive instruments across tasks.
- Supermodularity provides conditions for instruments to move together.

- Association and affiliation provide conditions for comparative statics to become cross-sectional covariation predictions.
- In industrial selling, hard-to-measure performance and important nonselling activities should push firms toward employment.
- Easy measurement and less important nonselling activities should push firms toward independent representation.
- Empirical evidence from industrial sales largely matches these predictions.
- Mixed evidence from franchising illustrates why agency models must be tailored to institutional details.

## 10 Conclusion

### 10.1 Core conclusion

Holmström and Milgrom argue that the firm should be understood as an incentive system. Employment and contracting differ across many margins because incentive instruments are jointly optimized. Compensation, asset ownership, monitoring, and job restrictions should be analysed as complements and substitutes inside one system.

### 10.2 Economic intuition

In multitask work, stronger incentives for measurable tasks can pull effort away from unmeasured tasks. The firm can respond by lowering performance pay, owning assets itself, restricting outside activities, and monitoring directly. These restrictions may look bureaucratic, but they can be efficient ways to balance incentives.

### 10.3 Incremental contribution in one sentence

The paper's distinctive contribution is to move the theory of the firm from isolated contractual attributes to a system-level theory of complementary incentive instruments, with a statistical framework for predicting their covariation in data.

### 10.4 Takeaway

Do not ask whether commissions, asset ownership, or authority individually explain the firm. Ask how they fit together. The firm is an institutional package for balancing incentives across multiple activities when market incentives would be too narrow or too strong.